Codebusters C

West-Windsor Plainsboro North CSE

02 August 2025



- You will have **50 minutes** to complete this test.
- You may bring up to **(3)** 4 or 5 function calculators, if you have any additional please place it underneath your seat
- You may use the space on the exam as scratch paper, but please make sure that your answers are clear. Either write your answers separately and box them, or write them directly over/under the ciphertext.
- You may take apart the test as you wish, however you should write your team number **on top of every page** and a **250 point deduction** will be taken if the pages are **not** put back in the correct order
- The first question on the exam is **Timed**. If you believe you have solved the question in the first 10 minutes, signal to the event supervisor who will come over to check it. If you are correct, your time will be recorded and a bonus issued; if you have more than 2 errors you will be notified and allowed to retry for a bonus throughout the first 10 minutes.
- If you successfully complete the timed question after 10 minutes, you may still get the base points for the question however it will not be checked and no bonus will be awarded.
- There are 3 questions marked for **Special Bonus**: 17, 19, 22
- Page 2 contains scoring information. Pages 3 and 4 contain the reference materials. Page 5 (front of the 3rd sheet) is the **Timed Question**.

Team Name: _____

Team Number: _____

Written By:

Claire Lin

Scoring Table

Exam Score

Question	Points	Score
1	379	
2	238	
3	241	
4	523	
5	237	
6	196	
7	252	
8	179	
9	498	
10	291	
11	473	
12	291	

Question	Points	Score
13	357	
14	197	
15	221	
16	186	
17	371	
18	454	
19	224	
20	219	
21	247	
22	328	
23	248	
24	208	

Question	Points	Score
25	234	
26	249	
27	278	
28	243	
29	352	
30	253	
31	223	
32	279	
33	246	
Total:	9415	

Timed Question

Question Score: Circle one: 250 0

Time: _____
minutes : seconds

Timed Bonus: $(600 -$ _____ $) \times 2 =$ _____
time in seconds timed bonus

Total TQ Score: _____ $+$ _____ $=$ _____
question score timed bonus total TQ score

Special Bonus

Circle one. Special bonus is only awarded for solving the question for full score. Special bonus questions: 17, 19, 22.

- 1 Correct = 150
- 2 Correct = 400
- 3 Correct = 750
- 0 Correct = 0

Total

Final Score = _____ $+$ _____ $+$ _____ $=$ _____
Exam Score Total TQ Total Special Bonus

Rank: _____

Tables

The tables below may be useful.

Keys	a	b	c	d	e	f	g	h	i	j	k	l	m
A,B	n	o	p	q	r	s	t	u	v	w	x	y	z
C,D	o	p	q	r	s	t	u	v	w	x	y	z	n
E,F	p	q	r	s	t	u	v	w	x	y	z	n	o
G,H	q	r	s	t	u	v	w	x	y	z	n	o	p
I,J	r	s	t	u	v	w	x	y	z	n	o	p	q
K,L	s	t	u	v	w	x	y	z	n	o	p	q	r
M,N	t	u	v	w	x	y	z	n	o	p	q	r	s
O,P	u	v	w	x	y	z	n	o	p	q	r	s	t
Q,R	v	w	x	y	z	n	o	p	q	r	s	t	u
S,T	w	x	y	z	n	o	p	q	r	s	t	u	v
U,V	x	y	z	n	o	p	q	r	s	t	u	v	w
W,X	y	z	n	o	p	q	r	s	t	u	v	w	x
Y,Z	z	n	o	p	q	r	s	t	u	v	w	x	y

A	AAAAA
B	AAAAB
C	AAABA
D	AAABB
E	AABAA
F	AABAB
G	AABBA
H	AABBB
I/J	ABAAA
K	ABAAB
L	ABABA
M	ABABB

N	ABBAA
O	ABBAB
P	ABBBA
Q	ABBBB
R	BAAAA
S	BAAAB
T	BAABA
U/V	BAABB
W	BABAA
X	BABAB
Y	BABBA
Z	BABBB

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Frequency of English Letters

E - 12.51%	S - 6.54%	C - 3.06%	G - 1.96%	K - 0.67%
T - 9.25%	R - 6.12%	U - 2.71%	W - 1.92%	X - 0.19%
A - 8.04%	H - 5.49%	M - 2.53%	Y - 1.73%	J - 0.16%
O - 7.60%	L - 4.14%	F - 2.30%	B - 1.54%	Q - 0.11%
I - 7.26%	D - 3.99%	P - 2.00%	V - 0.99%	Z - 0.09%
N - 7.09%				

Frequency of Spanish Letters

E - 14.08%	I - 5.98%	M - 3.08%	Y - 1.09%	Z - 0.47%
A - 12.16%	L - 5.24%	P - 2.89%	V - 1.05%	Ñ - 0.17%
O - 9.20%	U - 4.69%	B - 1.49%	G - 1.00%	X - 0.14%
S - 7.20%	D - 4.67%	H - 1.18%	F - 0.69%	K - 0.11%
N - 6.83%	T - 4.60%	Q - 1.11%	J - 0.52%	W - 0.04%
R - 6.41%	C - 3.87%			

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, "burro" is considered as five letters: "b-u-r-r-o" and not as four letters "b-u-rr-o."

Morse Code

A	●−
B	−●●●
C	−●−●
D	−●●
E	●

F	●●−●
G	––●
H	●●●●
I	●●
J	●–––

K	−●−
L	●−●●
M	––
N	−●
O	–––

P	●––●
Q	––●−
R	●−●
S	●●●
T	−

U	●●−
V	●●●−
W	●––
X	−●●−
Y	−●––
Z	––●●

0	–––––
1	●––––

2	●●–––
3	●●●––

4	●●●●−
5	●●●●●

6	−●●●●
7	––●●●

8	–––●●
9	––––●

●	E
●●	I
●●●	S
●●●●	H
−●●●	B
–––––	0
●●●●●	5

−	T
●−	A
●●−	U
●●●−	V
−●●−	X
●––––	1
−●●●●	6

−●	N
●−●	R
●●−●	F
−●−●	C
●●–––	2
––●●●	7

––	M
●––	W
●−●●	L
−●––	Y
●●●––	3
–––●●	8

−●●	D
●––●	P
––●●	Z
●●●●−	4
––––●	9

−●−	K
●–––	J
––●−	Q

––●	G
–––	O

Timed Question. [250 Points] Solve this **Aristocrat**. When you have finished, **yell out your favorite month** so your answer can be checked and time recorded.

BCOZYSYN UP JZSDC SG O IZCCJPH, XQU UVCY SU UQZYG

SYUP O XQZJCY TVCY RPQ XCLPHC O LVOQIICQZ.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	1	10	1	0	0	2	2	3	3	0	2	0	1	5	5	6	1	5	1	6	3	0	3	7	6
Replacement																										

1. (379 points) Solve this **K1 Xenocrypt** about my inability to write in Spanish.

IHUHJGEJ SRIHG HUKGEDEG VRIR HUVH HZJPHQ UEQ

EQVHGQHV, SHGR QR UH HUSJCRO, JUE TWH QHKHUEVH

FRRFOH VGJQUOJVH SJGJ JAWIJGPH.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	0	1	1	7	2	9	20	4	11	2	0	0	0	0	3	2	7	8	4	1	9	7	2	0	0	1
Replacement																											

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2. (238 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains the word **HOT**.

IZOSS TITKT RWHNE AOALX COTEM TCPGX SHIEW DAONX

TYTBA ENCIX AHAIL RSROX

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3. (241 points) Solve this **Aristocrat** about pollution.

UI IFFA UI YB OLFIIBS QPB VFLSBL, JC IDIQBL

BHOEUDJBS, "EFFT! QPB LDKBL DI EDRPQ VEZB!" UI DW

IPB PUIA'Q IBBA YUQBL VBWFLB.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	16	1	6	4	7	0	1	12	2	1	8	0	0	2	5	6	1	3	1	6	3	2	0	2	1
Replacement																										

4. (523 points) Solve this **K2 Patristocrat** containing some general advice.

NEHLH QRDDW DQWUM XHMIF HXIZU QEINH DDMUI ORNRM

RFJIM MRXDH XONGH PHLDH NNEWN ZLWTU IOZIQ G

Replacement																										
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	0	0	7	3	2	2	9	7	1	0	3	6	7	3	1	4	5	0	1	4	0	4	4	0	3

5. (237 points) Solve this **K1 Aristocrat** about friendship. You are told that the keyword used is 8 letters long. What is the keyword? **box** your final answer.

OTP VNZUAKD UZ RAZRPKO QXLRW SXAAMPM OTP DPZUAKD UZ

ZLEH QXNP KAQPD, PYQKLRUZJ PLRT AOTPK SAK OTP XLDO

OUYP.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	8	0	0	5	1	0	0	1	0	1	7	5	2	2	7	12	4	5	2	5	6	1	1	4	2	7
Replacement																										

6. (196 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **FALL**.

$$\begin{pmatrix} F & A \\ L & L \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 11 & 11 \end{pmatrix}$$

DJAXUIAFUSDWUSPLZDDIOXED

7. (252 points) Solve this **Aristocrat** about dedication.

VE CUFDQUZD FZQBNVF FDPCBKD-NDWIBDBF. DWB DPQT WNF

RBBK TPII MT NDWIBDBF XBDDUKX UK DWBUQ AMQOMPDF

NTDBQ FPKFBD.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	13	2	16	1	10	0	0	4	0	5	0	3	5	1	5	6	1	0	4	5	2	5	2	0	2
Replacement																										

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8. (179 points) Decode this phrase that was encoded using the **Checkerboard** cipher with a polybius keyword of **DESSERT**.

RE ES AW RE AW EE RS TT TW EW TE TE AW ET AT

RE ES RW TW TT TT TW TE TT ES RS AS RE RS AS

TS TE

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9. (498 points) Solve this **Patristocrat** about screenagers who won't get off their phones.

KXUOV PDVPX ZGQFG KGPXU PMIVB VPDIV TGVKH PGHTX

ZGZUF MGKXX ZVPDK XHMHT HFHAF DGPGF UXVHP

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	1	0	4	0	5	9	7	2	0	5	0	3	0	1	9	1	0	0	3	4	8	0	8	0	4
Replacement																										

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10. (291 points) Decode this phrase that was encoded using the **Fractionated Morse** cipher. You are told the plaintext ends with **YEAR** and **0 Xs of padding** at the very end.

N L X R C Z D T A B R Q F K Q I

Q K W K Q K V R Z S W M B T T C

Z E L S Y A E C E L D Q F

E C K W N S Z Q A A R Z S U B D

Replacement																										
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

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11. (473 points) Solve this **K1 Patristocrat** about spending time with friends.

BTLQL OBBKX LBYTP MSYCB KONCJ KMSPJ PMNYX OZYMB

PMLYC OXLLB CZYMP OCXXL JELLV NPGMK STB

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	8	5	0	1	0	1	0	0	3	4	9	7	3	5	6	1	0	3	3	0	1	0	5	6	2
Replacement																										

12. (291 points) Solve this **Aristocrat** about a well-known sports shop that was redacted.

KMEJASBV KZEMK ZPCI P VJIPA CPJSIAU EY IFGSMXIBA

YEJ ICIJU KMEJA ILWIMA KHSXXSBV. HZU PJI AZIJI EBNU

YNEPASIK PBT OST VEVVNIK?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	8	5	3	0	8	1	1	2	14	8	7	1	5	3	1	7	0	0	7	2	4	6	1	3	3	4
Replacement																										

13. (357 points) Decode this phrase that was encoded using the **Fractionated Morse** cipher. You are told the plaintext contains **MOST** corresponding to **JHPAY**.

R Q U E Q F U F G Y Q Q E C V J

P P B H V S W K T Z P U B

J H P A Y N F C S T Q E L S C

Y R Q V U U K U

Replacement																											
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

14. (197 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **SUMMER**.

RPTHJW FNFMEQ UVNTDK YRIINQ ETBRJQ JOHWOD YEHWMT

NDVXD

15. (221 points) Solve this **cryptarithm** for $80274\ 91042$. Write out your final answer and box it.

```
Base 10 Addition
Answer: 80274 91042 (two words)

     GOLD
+   HORSE
---------
   EASTER
```

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16. (186 points) Decode this phrase that was encoded using the **Checkerboard** cipher with a polybius keyword of **VISION**.

```
OR KR SR OE SR    OE WA OS WE OR    WA WR RS KE OA

WR WE KE OS OA    KE WA WT WA OS    OT OS OR KT OS

WT KR WT WE RT
```


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17. (371 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher. You are told that the keyword used to encode it is between 3 and 7 letters long and that the numbers 28 58 67 36 57 (third line) decode to **OTHER**. ★***This question is a special bonus question.***

73 47 59 27 84 55 38 48 86 44 30 26 84 28 46 48

96 26 60 56 55 47 27 68 85 36 49 57 65 26 26 28

73 55 38 36 86 44 28 48 73 56 28 58 67 36 57 36

57 55 58

	1	2	3	4	5
1					
2					
3					
4					
5					

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18. (454 points) Solve this **K2 Xenocrypt** about my sister's habit.

TS YZJTUAU KSEMZ JCVUAXC LCXUK TSK KCHMWSCAZK XZ

WCALUWLC DCJFMZ HZ XU TMWYU DZJZRU SJ UH KCLUAC.

Replacement																											
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	0	10	2	1	1	0	3	0	5	6	4	4	0	0	0	0	0	1	5	4	10	1	4	4	2	8

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19. (224 points) Decode this phrase that was encoded using the **Checkerboard** cipher. You are told that ciphertext **PI LI LT NH LH** (Characters 12-16) decode to THING. ★***This question is a special bonus question.***

ST PI LR ND PI LT NH LH LR NH SR PI LI LT NH LH

LT ST LR AD PR LR SR ST PI LI PD LI LR ND AH PD

ST PI NI LR ND PI

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20. (219 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher with a keyword of **MEMORIES** and a polybius key of **HILARIOUS**.

84 47 55 63 50 35 46 38 73 77 52 53 39 44 55

68 93 46 95 66 36 35 64 55 86 43 73 66 30 34

64 46 53 63 73 42 48 66 53 45 56 65 56 33 47 54 63 46

	1	2	3	4	5
1					
2					
3					
4					
5					

---

21. (247 points) Solve this **Aristocrat** about college dorms.

FYU QCQSJCMQYUV MJU QCARKL ZJCA FYU ZPCCJ. KUNUJ

ARKW FYUH'JU ZMPPRKL ZJCA FYU QURPRKL! XPUMVU VUKW

YUPX.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	0	7	0	0	4	0	1	0	7	6	3	4	1	0	6	5	5	1	0	14	3	2	2	6	4
Replacement																										

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22. (328 points) Solve this **cryptarithm** for $4700\ 38140768$. Write out your final answer and box it. ★***This question is a special bonus question.***

Base 10 Multiplication
Answer: 4700 38140768 (two words)

```
   CLAIRE
x     BUM
---------
   ITBBME
 CCMAEBE
 BAIECE
---------
 MEALTIME
```

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23. (248 points) Decode this phrase that was encoded using the **Porta** cipher. You are told the plaintext ends with SHOES.

KUXEH RVNEO ENRWD VHUOW YWHMS LJLER CSEZW MRAAG

NNCBB DFZFN F

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24. (208 points) Decode this phrase that was encoded using the **Baconian** cipher. You are told that the plaintext starts with ends with RE.

BEBEBEABREBAERBEABEBREBAERBEABEBEBEBREBAEBREBAEBERBAEBE

RBEBAERABREARBEBEABEBRAREBEABERBEBEABEBREARBAERABERBEAB

ERABREABEBERABERBEBAERARBEABEBREBEBEBAEBREBAEBERARBEABE

BERBAREBARAEBREABERARBEBEBEABE

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25. (234 points) Solve this **Aristocrat** about graduation.

ZBC EKSZZG'E JKJVPGGJ LCZBV WCZBLSY YSP JBHOPUKP YZ

YPJCE MOYS YSPOC CPUHOYOZU ZX 'CODPCE JUH CZJHE' JY

LCJHBJYOZU.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	5	10	1	5	0	3	5	0	10	3	3	1	0	7	8	0	0	5	0	5	2	1	1	9	10
Replacement																										

26. (249 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **HEADPHONE**.

$$\begin{pmatrix} H & E & A \\ D & P & H \\ O & N & E \end{pmatrix} = \begin{pmatrix} 7 & 4 & 0 \\ 3 & 15 & 7 \\ 14 & 13 & 4 \end{pmatrix} \qquad \begin{pmatrix} 7 & 4 & 0 \\ 3 & 15 & 7 \\ 14 & 13 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} 19 & 14 & 8 \\ 6 & 8 & 25 \\ 5 & 3 & 21 \end{pmatrix}$$

PXOZBOWVGKOLCHKXSEOWQUYUXIC

27. (278 points) Solve this **Aristocrat** about a recent tournament in the US.

VJI VGIMVD VGIMVD-PNFI XAOT GBZAR XOE GNAA SB RBGM

NM PBBVTLAA JNQVBZD. DBO XLM'V XBMFNMXI KI

BVJIZGNQI.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	6	9	0	4	1	2	6	0	8	3	1	2	7	6	3	2	2	2	1	2	0	9	0	5	0	3
Replacement																										

28. (243 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **LEVEL**.

RNEVL NPINB EIHEI IDNII OAOLH FIGRE FGTEE GONCL P

MTLH VOOSD RTPWA OSIEX

29. (352 points) Solve this **K3 Aristocrat** about funny moments. You are told that the keyword used is 9 letters long. What is the keyword? box your final answer.

AWMAGW LWWV OM ANGG IMKW AKJLHP ML OEWFK XKFWLVP

QWTJNPW YMMV MLWP SFGG QW KWIWIQWKWV XMKWUWK.

K3	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	0	0	0	1	3	5	1	3	2	8	5	8	2	2	4	3	0	1	1	1	4	17	2	1	0
Replacement																										

30. (253 points) Solve this **K2 Aristocrat** about common misconception. You are told that the keyword used is 7 letters long. What is the keyword? box your final answer.

WHXD FOQXNR VYYU LHRQLE FOHX FOLD HJFCHVVD HEL

ILJHCRL DYC'EL AHFJOQXN LBZLEFR KY FOLQE FOQXN

Replacement																										
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	1	3	4	5	8	0	8	1	3	1	10	0	3	6	0	5	4	0	0	1	3	1	5	4	1

31. (223 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **GAMES**.

```
WRISG DNMRN TTOEL TTTOA MNUUH TGESG BMNNS SRNEI S

TTNN HANTA EIOEC TOGDR EEXPI HRSYO II
```

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32. (279 points) Decode this phrase that was encoded using the **Baconian** cipher. You are told that the plaintext starts with **UNHINGE**.

```
STARE HERON WHERE GUANO WEEDY GAMUT FOUNT TONER PREEN

NORTH PEENS DERBY GUARD TAUNT DONNA MOLAL ALEPH MOOCH

FACES HOUND HOMOS KNOCK FERAL ARROW DEVIL NOBLY SHIPS

SPIED TURBO LARVA TURDS TUMMY ABORT DOUGH COVEY ABETS

ADAGE YIELD GLINT ELAND PEENS RANKS TRULY VIVRE
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33. (246 points) Solve this **K2 Aristocrat** about traveling. You are told that the keyword used is 5 letters long. What is the keyword? box your final answer.

ESJB UN GTCI PJDTIUBO YTMO TP BIJZNRTIBJBUTZ? U

MTZ'B XUWO LJIN KOLJCNO U QT ZCYK PITY NUBBUZQ PTI

STCIN.

Replacement																									
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	8	4	1	1	0	1	0	8	6	2	2	2	6	5	4	2	1	2	11	8	0	1	1	3	5